

# Zuodong, WANG

## 王作栋

Ningbo, China  
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## RESEARCH INTERESTS

### Numerical Analysis

Hyperbolic conservation laws, fluid dynamics, semi-linear parabolic problems, PDEs with stiff source term, structure-preserving schemes, a posteriori error analysis

### Programming

C++, Python, Matlab, Julia

## ACADEMIC APPOINTMENTS

### Postdoctoral Researcher (2025-present)

Eastern institute of technology, Ningbo [link](#)

- Project: Efficient high-order structure-preserving schemes for evolution problems
- Advisor: Jie Shen

## EDUCATION

### PhD (2025)

CERMICS and SERENA team,

Ecole nationale des Ponts et Chaussées, Institut Polytechnique de Paris & Inria, Paris

- Thesis title: Efficient numerical schemes for evolution equations with singularities and shocks
- Defense date: 09/23/2025
- Advisors: Zhaonan Dong, Alexandre Ern
- Jury members: Gabriel Barrenechea, Roland Becker, Claire Chainais-Hillairet, Konstantinos Chrysafinos, Bruno Després, Zhaonan Dong, Alexandre Ern, Raphaële Herbin

### Master (2022)

Ecole nationale des Ponts et Chaussées, Paris-Est & Sorbonne Université

- Applied Mathematics: Modeling, Analysis and Simulation
- Grade: 16.10/20 (top class)

### Engineer's degree (2021-2022)

Ecole nationale des Ponts et Chaussées, department IMI

### Bachelor (2020)

Beijing University of Technology

- Mathematics and Applied Mathematics
- GPA: 3.77/4.0

## LANGUAGES

Chinese (Mandarin): Native,

English: Fluent, French: Fluent, Japanese: Duolingo

## CONFERENCES

### 11th International Conference on Computational Methods in Applied Mathematics

Vienna, Austria, 2026 – *invited talk*

### Tsinghua Sanya International Mathematics Forum on Advanced numerical schemes for multiphysics problems

Sanya, China, 2025 – *invited talk*

### 30th Biennial Conference on Numerical Analysis

Glasgow, UK, 2025 – *contributed talk*

### International Conference on Applied Mathematics

Hongkong, China, 2024 – *contributed talk*

### Finite Element Rodeo

Houston, USA, 2024 – *talk in mini-symposium*

### The 6th SIAM Texas-Louisiana Sectional Meeting

Lafayette, USA, 2023 – *talk in mini-symposium*

## TEACHING

### Numerical analysis (French)

2025 – *Paris Dauphine University*

### Explorations in Mathematics (English)

2024 – *Texas A&M university*

## AWARDS

### CSC Scholarship

Paris, 2020 – 2022

### Outstanding Graduate

Beijing University of Technology, 2020

### Model Student of Academic Records

Beijing University of Technology, 2016 – 2020

## HOBBIES

ACG(Anime, Comics, Games)

## EXPERIENCE

### Visiting scholar

Texas A&M university

*College station, USA, September 2023 – May 2024*

- Project: Design, analyse and testing Invariant-domain-preserving schemes for hyperbolic problems

### Internship (stage court)

Inria Centre de Paris

*Paris, France, June – September 2021*

- Project: Hybrid high-order method for elliptic PDEs on curved and complicated domains

## PUBLICATIONS

**Pressure-robust hp-a posteriori error estimates of  $H(\text{div})$ -conforming discontinuous Galerkin methods for the Stokes equations**

2026, Zhaonan Dong, Zuodong Wang, Lina Zhao – *Submitted* [link](#)

**A high-order nodally bound-preserving and mass-conservative method for linear fourth-order elliptic problems and its applications to nonlinear parabolic equations**

2026, Jie Shen, Zuodong Wang – *Submitted* [link](#)

**A posteriori error analysis and adaptivity of a space-time finite element method for the wave equation in second order formulation**

2026, Zhaonan Dong, Emmanuil H Georgoulis, Lorenzo Mascotto, Zuodong Wang – *Accepted in Numerische Mathematik* [link](#)

**A priori and a posteriori error estimates of a  $C^0$ -in-time method for the wave equation in second order formulation**

2026, Zhaonan Dong, Lorenzo Mascotto, Zuodong Wang – *Numerische Mathematik* [link](#)

**A bound-preserving scheme for the Allen-Cahn equation**

2025, Alexandre Ern, Zuodong Wang – *Computers & Mathematics with Applications* [link](#)

**Mass conservative limiting and applications to the approximation of the steady-state radiation transport equations**

2025, Jean-Luc Guermond, Zuodong Wang – *Journal of Computational Physics* [link](#)

**Asymptotic and invariant-domain preserving schemes for scalar conservation equations with stiff source terms and multiple equilibrium points**

2024, Alexandre Ern, Jean-Luc Guermond, Zuodong Wang – *Journal of Scientific Computing* [link](#)

**Hybrid high-order methods for elliptic PDEs on curved and complicated domains**

2022, Zhaonan, Dong, Zuodong Wang – *13th ICOSAHOM 2020/2021* [link](#)

**The Proof of 2n Circle Arrangement Conjecture**

2017, Zuodong Wang – *Pure Mathematics* [link](#)